ORIUS in One View

A Safety Layer for Physical AI

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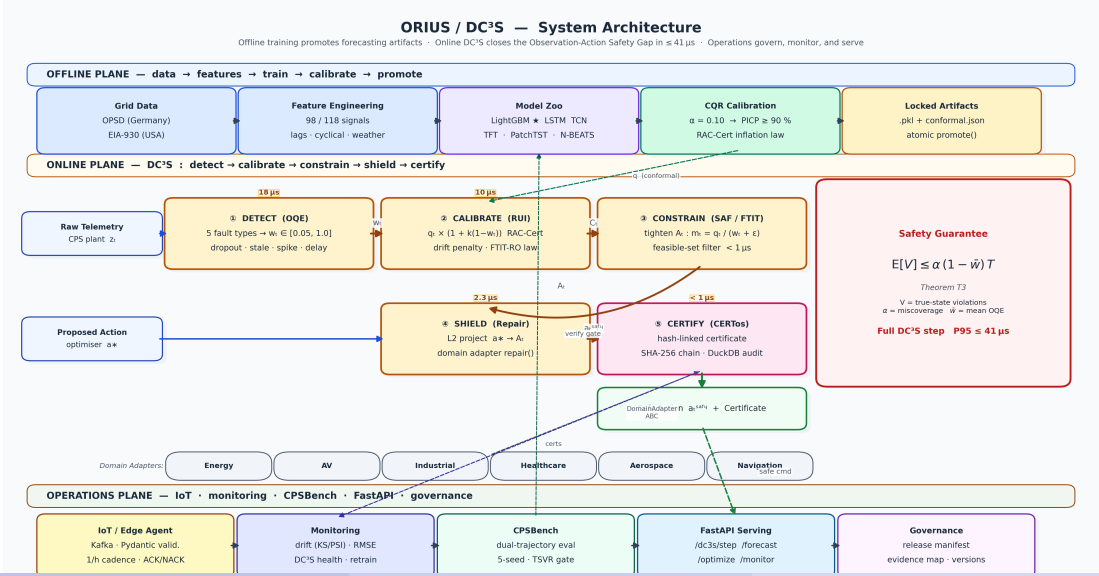
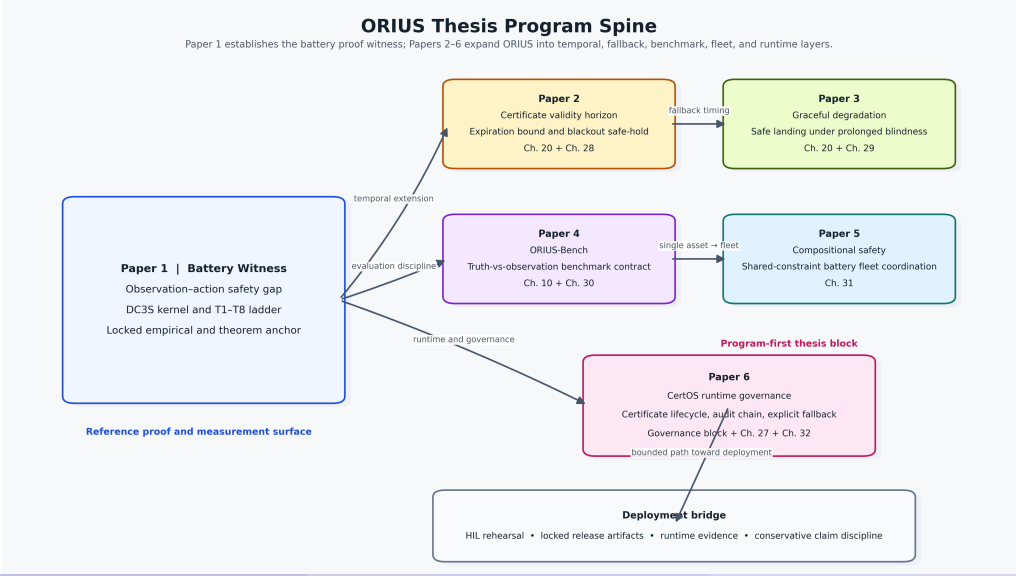
Why ORIUS Exists

- Physical AI can look safe in observed state while violating the true state of the plant.
- ORIUS is built to close that hidden observability gap online.
- The core runtime loop is Detect, Calibrate, Constrain, Shield, and Certify.

Architecture



How the Book Is Organized



Six-Domain Scope

ORIUS Theory-to-Runtime-to-Domain Flow

One universal safety argument, six domain instantiations, one parity gate

Current domain status

Universal-first book claim: ORIUS is one safety architecture with one runtime contract. Equal-domain rhetoric is governed by the parity gate: battery is the reference witness; industrial, healthcare, and bounded AV are defended; navigation and aerospace remain explicitly open.

Current Evidence Status

ORIUS Equal-Domain Parity Gate
Universal-first structure with explicit evidence asymmetry

Domain	Raw	Train	Replay	Soundness	Fallback	CertOS	P5	Tier	Current Blocker
Battery Energy Storage	locked reference	pass	pass	reference witness	safe hold validated	evaluated	evaluated	reference	battery reference row
Autonomous Vehicles	verified	pass	pass	pass under ttc entry barrier contract	bounded brake fallback validated	evaluated	gated pending shared constraint surface	proof validated	multi lane and higher dimensional repair open
Industrial Process Control	verified	pass	pass	pass	bounded runtime pass	evaluated	evaluated	proof validated	bounded to current plant family
Medical / Healthcare Monitoring	verified	pass	pass	pass	bounded runtime pass	evaluated	gated pending shared constraint surface	proof validated	bounded to current monitoring and intervention contract
Navigation / Guidance	blocked real data gap	blocked real data gap	portability only	blocked real data gap	bounded runtime pass	gated	gated	shadow synthetic	navigation real data row missing
Aerospace Control	placeholder surface	placeholder surface	experimental replay only	experimental placeholder	bounded runtime pass	gated	gated	experimental	real multi flight safety task missing

Green = defended or evaluated, Blue = reference witness, Amber = bounded/gated, Red = open blocker.

What the Monograph Can Claim Today

- Battery is the deepest reference witness.
- Industrial, healthcare, and bounded AV rows are defended.
- Navigation and aerospace are still explicit open gates.
- The book is universal-first in structure and conservative in evidence.

Deliverables

- 455-page book-length PDF
- 183-reference bibliography
- five-reviewer R1-style dossier
- parity matrix and governed claim surface